Title : FloodCast

problem statement: India loses ₹50,000+ crores annually due to flooding. Farmers, residents, and disaster management officials often receive flood warnings too late or after heavy rainfall has already begun. Current weather data is fragmented across multiple sources, making it difficult to predict and prepare effectively.

There is a clear need for a centralized, predictive system that can forecast flood risk 7–10 days in advance by integrating rainfall and terrain data. Such a system would empower communities and officials to take proactive measures, reducing loss of life, property, and agricultural output.

Solution

The proposed solution is a **centralized predictive system** called "FloodCast". It integrates **rainfall, terrain, and weather forecast data** from multiple sources into one platform. Using **data analytics and visualization**, the system will:

1. Forecast Flood Risk (7–10 days ahead)

2. Provide an Interactive Dashboard:

- District-wise flood risk heatmap.
- Rainfall & terrain trend charts.
- Prediction panel with probability scores.
- Alerts/notifications for high-risk zones.

3.**Support Decision-Making for Users:**

- Farmers → plan sowing/harvesting, protect crops.
- Residents → prepare for evacuation and safeguard property.
- Disaster officials → allocate resources and plan rescue routes.
- Insurance companies → assess claims and predict risk zones.

End Users

1. Farmers
2. Residents (local communities)
2. Disaster Management Officials
3. Insurance Companies

Message to Users

Our system gives you early flood risk forecasts 7 – 10 days in advance*, so you can protect your crops, safeguard your family, and plan rescue operations before disaster strikes. Stay informed, stay safe, stay prepared.″*

Project Dashboard :

